

Extending value stream mapping through waste definition beyond customer perspective

Mahvish Khurum, Kai Petersen^{*,†} and Tony Gorschek

School of Computing, Blekinge Institute of Technology, Box 520, SE-372 25, Karlskrona, Sweden

ABSTRACT

Value stream mapping (VSM) is one of the several Lean practices, which has recently attracted interest in the software engineering community. In other contexts (such as military, health and production), VSM has achieved considerable improvements in processes and products. The goal is to capitalize on these benefits in the software intensive product development context. The primary contribution is that we are extending the definition of waste to fit in the software intensive product development context. As traditionally in VSM everything that is not considered valuable is waste, we do this practically by looking at value beyond the customer perspective and using the software value map. An evaluation has been conducted through an industrial case study. First, the instantiation and motivations for selecting certain strategies have been provided. Second, the outcome of the VSM is described in detail. The instantiation of VSM via workshops was considered good as workshops allowed active interaction and discussion stakeholders' groups that are distant from each other in the regular work. With respect to waste and improvement identification, the participants were able to identify similar improvement suggestions. In a retrospective, the value stream approach was perceived positively by the practitioners with respect to process and outcome. Copyright © 2014 John Wiley & Sons, Ltd.

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KEY WORDS: lean software development; software process improvement; value stream mapping; software product management; software project management

1. INTRODUCTION

Software development industry is faced with the challenges such as global competitiveness and shorter time-to-market. In order to address these challenges, there is a need for better processes/tools/methods/models/techniques to create quality software quickly, repeatedly, and reliably. Lean initiatives in manufacturing, logistics, services, and product development have led to radical improvements in quality, cost and delivery time [1–5]; however, the question in hand is whether lean principles can possibly be as effective for software development? According to the authors in [5–7], the answer seems to be *of course*. Lean is emerging as a popular way to make processes better [5]. Lean is a way of thinking that supports organizations to '*specify value, line up value-creating actions in the best sequence, conduct these activities without interruption whenever someone requests them, and perform them more and more effectively*' [8].

According to Wang *et al.* [9], the current understanding of lean software development is largely shaped by practitioners' writings [5, 10, 11]. The main focus of lean is the identification and elimination of waste from the process with respect to customer value [9]. Value stream mapping

^{*}Correspondence to: Kai Petersen, School of Computing, Blekinge Institute of Technology, Box 520, SE-372 25, Karlskrona, Sweden.

[†]E-mail: Kai.Petersen@bth.se

(VSM) is a one of the several lean practices that can be employed to eliminate waste and optimize the whole [5]. VSM assists to identify bottlenecks in a process that impedes it from flowing at its optimum [5, 12]. It enables organization to understand any workflow with an end-to-end perspective [12–14]. VSM has been applied at a relatively smaller scale with success in the context of manufacturing, where it enabled substantial reduction in lead time, setup time, cycle time and inventory levels [15]. Another example is the application of VSM to supplier networks, which enabled efficiency gains in collaborating with suppliers, and provided valuable input for benchmarking [16]. Besides the benefits related to concrete outcomes, the use of VSM helps to visualize the process seeing the end-to-end flow, guides to focus on not only waste, but also value, facilitates the use of a common language, guides scheduling and forecasting activities, and forms the basis for lean production (cf. [17]). Overall, the outcomes for value stream studies are positive, and hence, software engineers should capitalize on the benefits. However, VSM is mainly focused on time, whereas value and waste in software engineering are more complex. VSM is not well investigated within software engineering ([18] claimed being the first empirical study); however, this study was performed using a simplified view.

The Massachusetts Institute of Technology (MIT) Product Development VSM (PDVSM) manual provides guidelines as how to do VSM in product development based on the authors' experiences in lean engineering within the aerospace industry [12]. The main focus of PDVSM manual is cost reduction with focus on value adding activities. Traditionally, waste is defined as any activity that consumes resources, time or space but does not add value to the product as perceived by the customer. All lean approaches focus on identifying and eliminating waste (see [19] for further details on the measurement and visualization of the flow of lean software development). However, most of the literature reporting use of VSM talks about the seven wastes purely in manufacturing context. Whereas, in the context of software development, there can be activities and steps that do not add value directly and thus may be termed as waste. For example, in manufacturing or engineering, performing detailed analysis to ensure architectural flexibility and maintainability for future sustainability might be termed as a waste. Whereas in the software development context, it could be highly value-adding for future additions and modifications consequently reducing lead-time for future releases. Moreover, it is fundamental to realize that a reduction in lead time might be negatively impacting other value considerations, such as quality of the product and architecture, human capital value, innovation capability and so on. Thus, in an effort to eliminate waste, extreme caution has to be taken. Value is the end (i.e., what we should deliver), and removing waste is the means to maximize value. To the best of our knowledge, this has not been discussed explicitly in existing literature related to VSM. Our primary contribution is that we are extending the definition of waste. As everything that is not considered valuable for customer is waste, we do this practically by looking at value beyond the customer perspective, and using the software value map [20] to drive waste identification. Detailed evaluation of VSM in an industrial case at Ericsson is a contribution within software product development context as such an extensive evaluation has not been reported in Software Engineering literature so far. A detailed illustration, via VSM application at Ericsson, demonstrates usability and usefulness of the proposed extension.

Given that there are too few applications in the software engineering context, the goal of this research is to evaluate the ability of VSM to identify waste and value in the software engineering context, which has been very valuable in other domains. By design, the practitioners will evaluate their processes from the additional value and waste perspectives. This study (a) illustrates the instantiation of the value stream framework in an explanatory way, allowing others to learn from the experience of executing the process; and (b) evaluates whether VSM supports the practitioners in identifying waste in the current situation and supports them to identify improvements consistently. These two contributions relate to the two separate sections in the results, namely instantiation of the framework (Section 7.1 relating to (a), and outcome of the framework application (Section 7.2 relating to (b)). Detailed experiences of (a) have not been reported yet, and for (b), no results exist so far, given that modifications have been made to the value stream process.

The remainder of the paper is structured as follows: Section 2 presents background on lean and VSM. Section 3 contains related work focusing on lean in general and VSM application in particular. Section 4 presents the software value map used to evaluate the impact of improvements on different value considerations. Section 5 presents the VSM framework for software development. Section 6 contains the case study design used to evaluate the VSM frame work. Section 7 illustrates the

instantiation of the framework in the case company, as well as the documented outcome of applying the framework. The implications and lessons learned are discussed in Section 8. Section 9 concludes the paper.

2. BACKGROUND

The concept of lean is not a concept unique to software development. Its origin can be traced back to the Toyota Production System in the 1950s [21]. However, Toyota revolutionized the automobile industry with their approach of ‘lean manufacturing’ in the 1980s. The core five inherently interlinked guiding lean concepts taken in [9] are the following:

- value: It is defined by the customer, and it is extremely important to have a clear understanding of what that is.
- value stream: A map that identifies every step in the process and classifies each step with respect to the value it adds.
- flow: It is important that the production process flows continuously.
- pull: Customer orders pull product, guaranteeing that nothing is built before it is actually needed.
- perfection: Striving toward perfection in the process by continuously identifying and eliminating waste.

In the early 2000s, Poppendieck and Poppendieck further transferred these principles from the manufacturing domain to specifically suit the software development context and called it ‘lean software development’ [5]. Table I lists the two sets of lean principles as stated in [9]. They overlap to a large extent reflecting the core and essence of lean approaches.

The application context of the VSM presented in this paper focuses on the product development activities associated with software intensive products. We recognize there are other important associated phases such as organizational issues [1]; however, this is out of the scope of this paper. It is necessary to highlight that the focus of the paper is not to compare/contrast lean software development with agile development practices, rather the objective is to propose extension to what qualifies as waste and enabling value-based evaluation of improvements in the context of software intensive product development. For a comprehensive discussion on the characteristics and differences between lean and agile, please see [22, 9].

3. RELATED WORK

3.1. Experiences with value stream mapping

Rother and Shook [17] provide a detailed explanation on the practical implementation of VSM, which is also used in services-related industries and other sectors. The ‘Lean Advancements Initiative’ research consortium at MIT has for many years applied lean concepts for improving aerospace product development processes. Their main work ‘PDVSM manual’ [12] offers extensive academic research for learning. The MIT manual uses *process activity mapping* as a tool for VSM. Pavnaskar and Gershenson have discussed some differences between manufacturing and engineering and have

Table I. Lean principles for software development (taken from [9]).

Lean principles by Poppendieck and Poppendieck [5]	Lean principles by Reinertsen [11]
Eliminate waste	Use an economic view
Build quality in	Manage queues
Create knowledge	Exploit variability
Defer commitment	Reduce batch size
Deliver fast	Apply Work in Progress constraints
Respect people	Control flow under uncertainty
Optimize the whole	Use fast feedback
	Decentralize control

proposed a set of icons, definition of process data, and added representation of parallel and sequencing processing [23]. There are six other tools for VSM, namely *production variety funnel*, *supply chain response matrix*, *demand amplification mapping*, *quality filter mapping*, *decision point analysis* and *physical structure mapping* [24]. However, they are more suitable for production and manufacturing industries given the type of waste to be identified and reduced/eliminated (for details refer to [13, 25, 26]).

Mujtaba *et al.* applied VSM to reduce lead time in software product customizations and provided some hints and tips as how different steps in 'VSM' (drawing current state map, identifying waste, etc.) can be performed [18]. However, this study only focused on reducing lead time and did not consider value considerations beyond customers' perspective.

3.2. Seven wastes in lean

The traditional seven wastes from lean manufacturing and their mapping the wastes in lean software development [5] are given in Table II.

3.3. Waste beyond the customer perspective

Defining the value of software product development work is an ongoing and probably never-ending endeavor [12]. As stressed in the MIT manual [12], it is fundamental to understand that without a working definition of the value created by the process being mapped, and an appreciation for how that value is created, one cannot guide an improvement initiative. According to the authors of MIT manual, value has to be understood in two different contexts: (1) the value of the process output to the company; and (2) the creation of value while carrying out the individual tasks that make up the process [12].

The MIT manual [12] suggests some metrics to measure time (e.g., cycle time and lead time) and costs (e.g., recurring and nonrecurring costs). Time and cost are important value considerations. However, these are only project-focused short-term value considerations, and product-focused or organization-focused long-term value considerations are largely missing in the MIT manual. For example, customer satisfaction is one of them. The MIT manual does talk about customer satisfaction; however, it does not break it down to what actually customer satisfaction means and how it can be evaluated. Chase proposed a list of aspects of value that a task could contribute [27]. However, a detailed account of value considerations relevant for different perspectives such as customer and internal business value are missing.

Table II. Wastes in manufacturing and their mapping to software development (taken from [22]).

Lean manufacturing	Software development
Inventory: intermediate work products and work in progress	W1: Partially performed work: work in progress that does not have a value until it is completed (e.g., a segment of code written but has not been tested)
Overproduction: the number of items produced is higher than the number of items demanded	W2: Extra features: functionality that has been implemented, but does not provide any value to the customers
Extra processing: extra work is produced due to, for example, poor setup of machines	W3: Extra processes: process steps that are not really needed (for example, writing documents that are needed)
Transportation: transport of intermediate work-products (may be due to poor layout of the production line)	W4: Handovers: Many handovers create overload.
Motion: machines and people being moved around rather than being used to create value	W5: Motion/task switching: People may have to move to identify knowledge or may have disturbances in their work due to other people moving.
Waiting: an idle machine is waiting for input	W6: Delays: There are delays during development, for example, waiting times within a development team (team members being idle).
Defects: fixing problems in the products	W7: Defects: fixing problems (bugs, defects and enhancements) in the products

Moreover, the authors argue that MIT recommendations to classify tasks as value adding and non-value-adding are inadequate; due to the fact that different perspectives for value evaluations are ignored. For example, a certain task can be adding pure business value (e.g., improving product architecture maintainability), which externally does not have visible value from the customer perspective. According to the MIT recommendations, it should be regarded as non-value-adding as it does not add to a product or process definition or reduce uncertainty in any apparent way. Whereas from the business sustainability perspective, it improves product architecture sustainability by making it maintainable. Therefore, this task should not be regarded as non-value-adding or pure waste. In traditional manufacturing, a task adds value directly to a process or product, whereas in a software development context, there can be several tasks that do not add direct the product or process value (as exemplified earlier); however, they do add value to the business by making the product more sustainable.

Section 4 gives a brief account of the software value map, which is proposed to be used for extending waste definition by incorporating value considerations beyond customer during VSM.

4. SOFTWARE VALUE MAP

The software value map [20] provides a consolidated view of the software value concept utilizing four major perspectives: the financial, the customer, the internal business process and the innovation and learning. The value aspects (VAs) and value components (VCs) contained in the map are collected through extensive review of economics, management and value-based software engineering literature. The value map offers a unified view of value, which can be used by professionals to develop a common understanding of value, and acting as a decision support to assure no value perspective is unintentionally overlooked when taking process improvement decisions. The map contains links between different VAs thus making interrelationships explicit.

The taxonomy used to categorize the perspectives for measuring value was inspired by balanced scorecard (BSC) [28, 29]. BSC can be defined as a set of measures that gives managers a quick but complete view of the business using four main perspectives, namely the financial, customer, internal process, and innovation and learning [28, 29]. BSC combines the financial perspective with other perspectives to provide an overall view of the business performance and was thus considered as a good base for the categorization in the creation of the software value map [20].

Figure 1 shows an excerpt from the software value map. The categorization of VAs and VCs from various perspectives can be seen in the figure. Looking at Figure 1, 'customer perspective' is one of the four value perspectives. Customer perspective contains two VAs as follows: 'VA1.1: Perceived value' and 'VA1.2: Customer lifetime value'. Each value aspect further contains one or more subvalue aspects or VCs. For example, VA1.1 contains three subvalue aspects and two VCs, namely 'SVA1.1.1: Intrinsic value', 'SVA1.1.2: Delivery process value', 'VC1.1.3: Network externalities', 'VC1.1.4: Complimentary value' and 'SVA1.1.5: User experience value'. Looking further down the structure into SVA1.1.1: Intrinsic value, it contains five measurable VCs as follows: 'VC1.1.1.1: Functionality', 'VC1.1.1.2: Reliability', 'VC1.1.1.3: Usability', 'VC1.1.1.4: Maintainability' and 'VC1.1.1.5: Portability'.

5. VALUE STREAM MAPPING PROCESS

In this section, the concept of VSM is briefly described. The steps involved in VSM are discussed in detail, and extensions proposed to existing VSM practice are also presented. Each step has activities within it.

Steps and activities within each step of the VSM framework are shown in Figure 2. Here, it can be seen that the additions proposed by this paper are clearly marked with a star (**), indicating the extension to the VSM. In the following sections, a brief account of each step and associated activities is given, and a detailed account of the goal, inputs, outputs and actions to be taken within each activity are given in Appendix A of the Supporting Information. However, for further details please see [12]. To the best of the researchers' knowledge, no one has proposed strategies to carry out the activities along with their evaluation with respect to accuracy, time and resources, and capability to handle complexity (see Appendices B and C).

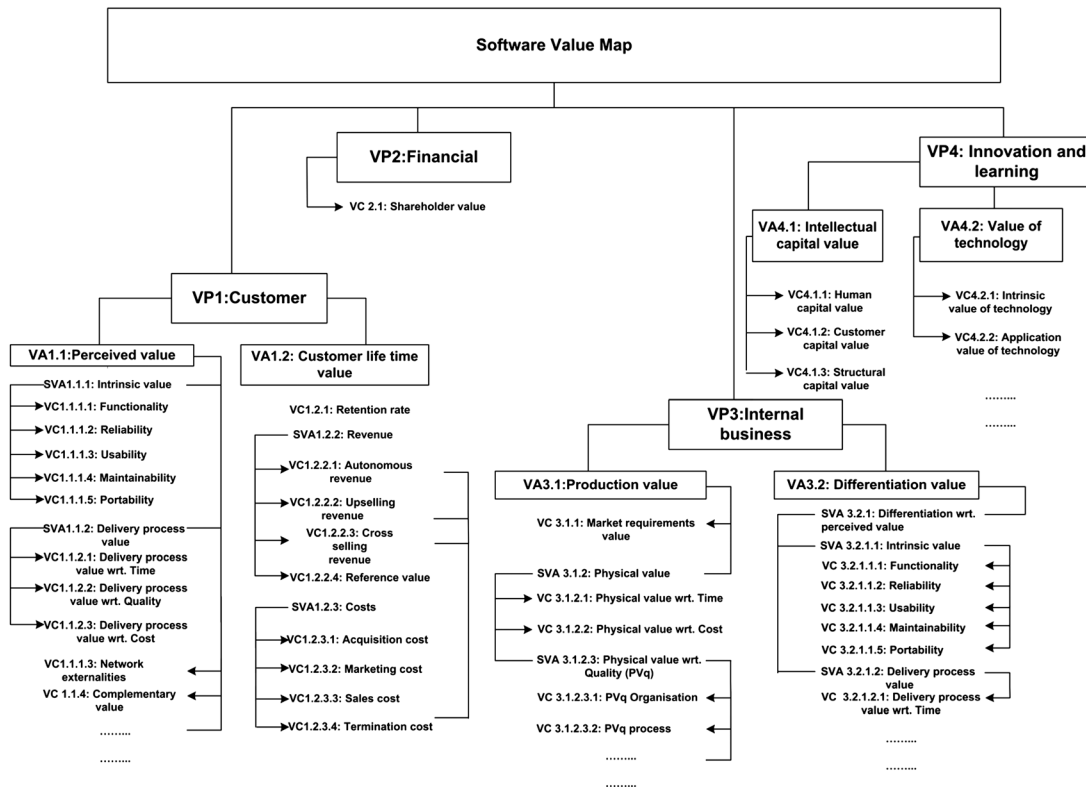


Figure 1. An excerpt from the software value map [20].

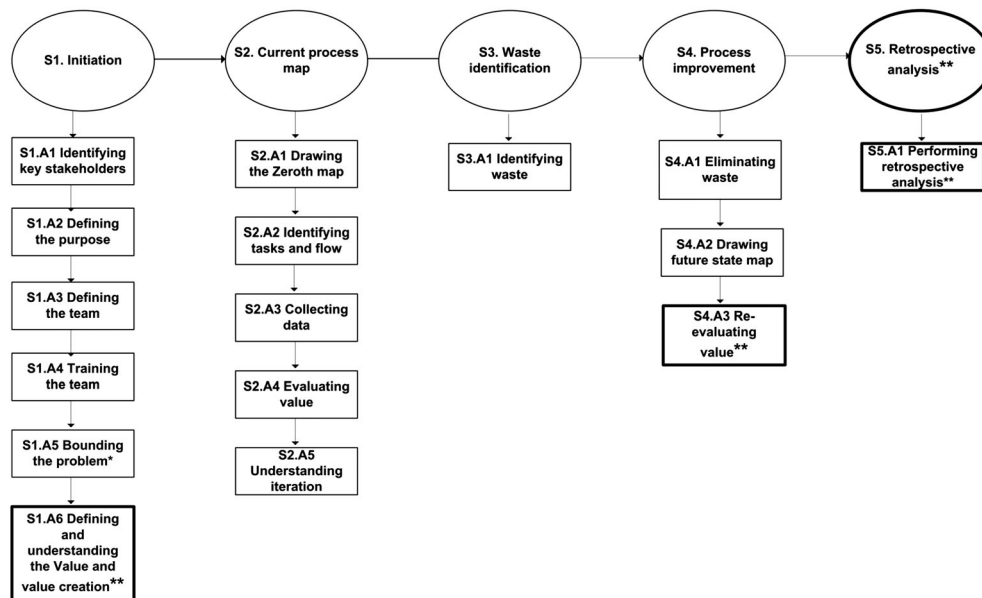


Figure 2. Value stream mapping process.

5.1. Notations used in value stream mapping

Figure 3 shows the notations used for drawing the VSM. These notations have been used by MIT manual [12] to demonstrate how VSM is applied to product development. A ‘process box’ specifies a single step or activity in the process in which the work is flowing. The step is written inside the box, for example, *specify requirement*. An ‘arrow’ specifies the direction of flow from one step to another. The ‘timeline’

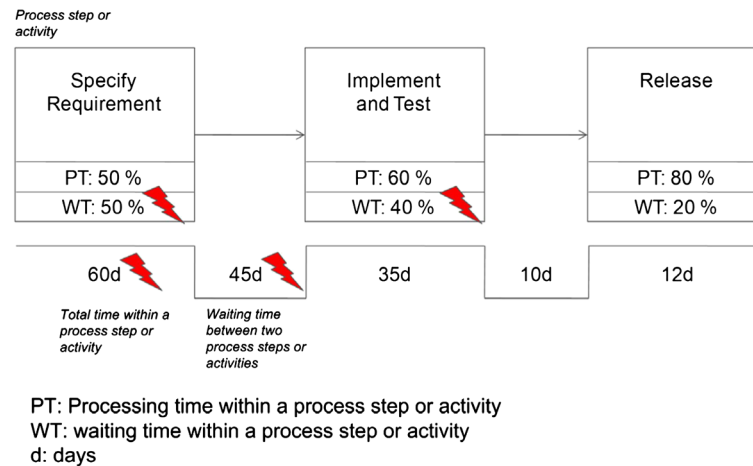


Figure 3. Value stream map notation.

specifies the sequence of process steps arranged in a chronological order with different times displayed along a line (drawn from left to right). The different times shown in Figure 3 represent different measurements, which are described below. The ‘burst signals’ represent alarm signals signifying waste. For example, long waiting times (WTs) with high improvement potential can be highlighted. Different measurements can be used in VSM (as detailed in [12]) but the ones shown in Figure 2 are related to time. *Processing time (PT)* is the time taken by one or several persons within the process step or activity to do the work. *WT* is the average wait time within a process step or activity or between subprocesses.

5.2. S1: Initiation

The first step includes all activities that are related to preparation and planning of the VSM activity.

S1.A1: Identifying key stakeholders

S1.A2: Defining the purpose

S1.A3: Defining the team

S1.A4: Training the team

S1.A5: Scoping the problem; and

S1.A6: Defining and understanding the value and value creation**: A detailed account of value considerations relevant for different perspectives such as customer and internal business value are missing. Thus in order to fill this gap, the proposal is to use the software value map (see Section 5) by Khurum *et al.* [20] for identifying value considerations relevant for the goal to be achieved (see S1.A6 in Appendix A).

5.3. S2: Current process map

In S2 of the VSM, the current state map is created. The map is the basis for identifying wastes and reasons for the occurrence of waste [12]. The most critical wastes will then be addressed through process improvement actions in S4.

S2.A1: Drawing the zeroth map

S2.A2: Identifying the tasks and flow

S2.A3: Collecting data

S2.A4: Evaluating value; and

S2.A5: Understanding iteration.

5.4. S3: Waste identification

S3.A1: Identifying waste: Once there is a clear vision of the process and value added by the tasks within the process, the next logical step is to proceed toward its improvement (step 4, see Figure 2).

Prior to that, a rudimentary step is identification of waste from the process in all of its forms. The sources of waste in product development processes are many and can be different, thus it is not possible to prescribe a recipe for its identification and elimination (for some recommendations, see S3.A1 in Appendix A).

5.5. S4: Process improvement

In this step, improvement alternatives are suggested, and the effect of the improvement alternatives are documented in the future state map.

S4.A1: Eliminating waste

S4.A2: Drawing future state map

S4.A3: Reevaluating value**: The MIT manual does not discuss reevaluation of value after drawing the future map, which the researchers believe is an essential step during VSM. The agreed strategies for waste reduction/elimination might be reducing/eliminating one type of waste, for example, WT; however, they might be compromising an important value consideration, for example, quality. Thus, it is important to evaluate the impact of the agreed strategies on the derived value considerations (derived from the goal, see S1.A3 in Appendix A). The impacts could be positive and/or negative on one or several VCs, and these need to be evaluated because cutting costs or reducing wait times are not the only important value considerations during software development.

5.6. S5: Retrospective analysis**

S5.A1: Performing retrospective analysis: After drawing current state map, identifying waste, drawing a future state map and reevaluation of value, it is deemed necessary to walk-through the future state maps and results of evaluation with the team members in a workshop. This has not been prescribed in the MIT manual, but the researchers emphasize it as a fundamental activity because it allows rectifying misunderstandings during the workshop and is similar to member checking in empirical studies (e.g., [30]). Furthermore, the retrospective should include a reflection on the value stream process to identify improvement opportunities. The next time the value stream is conducted, it can be performed in a more efficient way based on the feedback.

6. RESEARCH DESIGN

The research method used in this paper is a case study. Case studies are particularly well-suited to study phenomena in a real-world environment with the researcher making observations and consulting multiple sources of evidence (e.g., documentation, interviews and observations) [30, 31]. Given that the approach is new, we need qualitative insights (i.e., rationales) and gain a deep understanding of how to approach VSM in an industrial context. Therefore, a case study has been chosen. Experiments would not have been suitable as they would not scale, and population-based studies would not provide qualitative insights and rationales for instantiation that we wanted to achieve. The case being studied is Ericsson AB, a leading telecommunication company. The company is ISO 9001:2000 certified.

6.1. Case description

The case and context are described, as this allows for generalizing the results to a specific context. Other companies in a similar context are likely to find the results transferable to their context [32]. The process used at the company follows a system of systems (SoS) approach.

Multiple definitions of SoS exist (cf. [33, 34]). All agree on that a system in a SoS shall be possible to be distributed alone. However, a set of systems also shall have the capability to operate together in a SoS. Different strategies of how to develop the SoS are characterized in [33], namely virtual, collaborative, acknowledged and directed. The SoS investigated in this study is directed. According to the definition in [33], a directed SoS is characterized by a well-specified purpose, the development is centrally managed, and that 'component systems maintain an ability to operate independently, but their normal operational mode is subordinated to the central managed purpose' [33]. Recently, the term has been

recognized within the field of software engineering, requiring that an SoS should have a number of the characteristics shown in Table III. The overall architecture of the SoS studied at Ericsson consists of 12 systems.

The process used at the company is shown in Figure 4. In the first step, the high level requirements (HL-Reqs) for the overall SoS are specified. Before the requirements are handed over to compound system development a so-called ‘Go’ decision is taken, meaning that development resources are allocated to the HL-Reqs. When the decision is positive, teams specify a detailed requirements specification, which is then handed over to the concerned system(s). The requirements are then implemented for a specific development system, and they are integrated (also called system level test of last system version – LSV test). The development is performed in sprints run by agile development teams (Agile Team Sprints in Figure 5). Each system can be integrated independently of another system, which provides them some degree of operational and managerial independence (Table III). However, the versions of two systems have to be compatible when the SoS is integrated (compound

Table III. System of systems approach characteristics (taken from [35]).

Characteristics	Case company
Operational independence	X
Managerial independence	X
Integration of system into system of systems (SoS)	X
SoS comprised of complex systems	X
System suppliers deliver systems for integration	X
Complete technical oversight of SoS and system supply	X

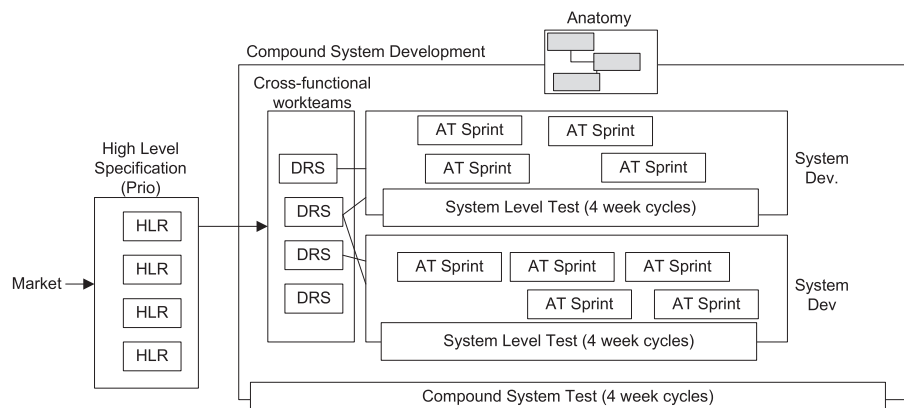


Figure 4. Systems of system development process at Ericsson.

Step 1: Getting Started	
S1.1 Key Stakeholders	[X] Direct internal customers [X] Indirect internal customers [] External customers
S1.2 Determining Purpose	[] Understand and communicate process [] Evaluate concrete improvement [X] Optimize towards specific target
S1.3 Defining the team	Strategic management: Chefs [X] Doers [] Systems: A/B Architecture and design: Chefs [X] Doers [] Systems: A/B Implementation: Chefs [X] Doers [] Systems: A/B Testing: Chefs [X] Doers [] Systems: A/B Delivery and Release: Chefs [X] Doers [] Systems: A/B
S1.4 Training the team	One hour training session on lean principles, purpose, and value stream notation and process
S1.5 Scoping the problem	[] Documentation analysis [] Interviews [X] Documentation analysis and discussion [] Workshop with team members
S1.6 Defining and understanding value	Value components considered: [X] Functionality [X] Quality of delivered product [X] Internal Efficiency [X] Delivery process value [X] Human capital value

Figure 5. Instantiation of the phase ‘getting started’.

system test C-LSV)). Each system is very complex, the largest system having more than 15 development teams. The size of the overall SoS measured in lines of code (LOC) is 5,000,000 LOC. This fulfills the characteristics of SoS development related to system complexity and integration. In order to make sure that the SoS is working together in the end, an overall system structure and design is developed, referred to as the anatomy. This allows having an oversight of the overall SoS, also making explicit how each system in the SoS contributes to the overall system goals. The company has been studied in earlier investigations related to the effect of moving from plan driven to agile, which provides deep insights to the reader, and these have been referenced as well [36, 37].

Looking at other context elements [32], the following details are added:

All systems are older than 5 years.

On principle level, the development process is incremental with projects adding increments to the code baseline on system and compound system level.

Within the teams and in the testing activities, agile practices are used, such as continuous integration, time boxing with sprints, face-to-face interaction (stand-up meetings, co-located teams), requirements prioritization with product backlogs, refactoring and system improvements.

The SoS is developed globally (with some development units being located; Sweden, India and China).

Two systems are mainly considered during the value stream activity, having a size of over 300,000 LOC each. However, because of confidentiality reasons, no further details about them can be revealed.

6.2. Research question

In this case study, two research questions are answered.

RQ1: What instantiations of the process do practitioners prefer?

There are alternatives of how to structure the value stream process (references to alternatives are provided throughout the framework description). Given that applications in software engineering are rare, we need to gain an understanding on how and why to instantiate in a certain way. A case study aimed at providing detailed reflections and hence qualitative (but deep) insights aid in contributing valuable evidence. According to the definition of Easterbrook *et al.*, this research question is a 'design question' [38].

RQ2: Is the instantiated process capable of identifying the current state (waste) and desired future state (improvement) accurately?

RQ2 relates to the key of value stream mapping: identifying wastes and improvements, as discussed in the related work (Section 4). This is achieved by presenting the outcome of the activity in detail. The presentation also aids practitioners in documenting their outcome. RQ2 would be characterized as a descriptive-process question according to Easterbrook *et al.* [38].

6.3. Data collection

Tailoring framework (generic solution) to context is a result and explanatory way of providing guidance and reasoning examples of how to do the tailoring, hence being considered a result of the case study itself. The decisions and reflections where made in the case context were performed in collaboration and discussions with the practitioners. Further case studies providing such detailed insights will help in further contributing to the understanding of how and why to do certain instantiations.

From a case study perspective, careful observation and documentation of the process and products of the value stream activity were required. For that purpose, two researchers were involved. One researcher was focused on moderating the workshop and providing guidance to the value stream map team. The other researcher was observing the VSM activities taking notes and documenting the process (e.g., by taking photographs, taking notes and storing them in a case study protocol). The

same was performed by a colleague working at Ericsson, who was also providing support in the moderation process.

Photographs: The photographs document the work products produced during the workshops, such as the zeroth maps, notes on whiteboards including yellow stickers/clusters of information structured together with the team.

Notes: The notes contain as much information as possible of the discussions taking place during the value stream activity. The researcher taking the notes handed these over to the moderator in order to complement them.

The data gathered during the value stream activity (utilizing the data collection approaches as presented in Section 8.1) provide an input to answer the research questions, in particular RQ2, which is concerned with the ability of accurately identifying waste and improvements.

6.4. Data analysis

The data analysis was based on the photographs and the notes in order to produce the report for the value stream map activity delivered to the company. This report is also the base for answering the research questions. The notes and information from the photographs were first clustered (e.g., for the current state map everything related to a high number of parallel items was put under a cluster 'number of parallel items'). The clustered information was tabulated and normatively described. The resulting report was reviewed by a fellow colleague at Ericsson who was also taking notes and took part in all activities of the value stream map. In addition, a walk-through of the report was conducted as a means of member checking [30] to make sure that the information documented during the workshops is accurate.

The effectiveness of the VSM process highly depends on the type of contribution. As the design question (RQ1) has been focused on understanding how and why to instantiate the framework in a certain way, the descriptions are qualitative of nature; hence, no quantitative measure is provided. The value lies in the rationales elicited. The second question RQ2 is about the capability to elicit wastes and values through the instantiated VSM accurately. By the nature of VSM, the nature of the outcome is highly dependent on the knowledge of experts providing their qualitative reflections. To determine whether these are accurate and insightful, three measures have been used that are considered strong empirical measures for the type of data collected. First, member checking [31] using walk-through has been utilized to evaluate whether we captured the information well; all persons participating in the value stream activity were involved. Second, we elicited improvements with two independent groups that were formed with similar roles working in the same context toward the same goal. As pointed out by Guest *et al.* [39] and Morse [40], saturation is an important measure to assess qualitative research. In this case, if both groups (given their homogeneity in set-up and goals) had produced very different results, it would have been an indication that the value stream activity is unfocused, unclear and poorly defined. However, if saturation is achieved (groups come to similar outcomes/conclusions), the accuracy of the outcome is high (cf. [40]). Third, a retrospective on the general perception of using VSM for process improvement has been collected from the practitioners.

6.5. Validity threats

Four types of validity threats are commonly distinguished, namely construct validity, internal validity, external validity and reliability [41–43].

Construct validity – obtaining the right measures for the concept being studied: The main risk in this context is that the researcher influences the outcome of the study with his/her presence. This is mainly related to trust. Given that the researcher who was moderating the workshop is also partially employed at the company, he/she was perceived as being internal. Hence, this threat to construct validity was reduced. Another risk is that the participants in the value stream activity misinterpret the intent and are not able to conduct the task in the best possible manner. To reduce this threat, trainings in lean and VSM were conducted. Given that there was very limited time for training, the threat was reduced, but not mitigated. During the workshop, the participants were always able to ask the

moderators for clarifications with respect to the task at hand, which also contributes positively to reducing the threat of misinterpretation.

External validity/generalizability – ability to generalize the usability and usefulness of the proposed framework: External validity/generalizability is a common threat in single case studies, as this is only able to focus on single cases, but at the same time allow gaining an in-depth understanding of the case and provide detailed insights of a real-world application. The results obtained in this case study are true in the context of the case. As argued by Wieringa in his recent keynote titled ‘Case study research in information systems engineering: How to generalize, how not to generalize, and how not to generalize too much’ at CAISE 2013, he argued that generalizing between cases should be performed by analogy and from one case to another [44]. Given that the research is qualitative in nature aiming at providing deep and detailed insights, as argued by Wieringa, random sampling-based approaches are not suitable for case study research. He rather argues that cases need to be compared by architectural analogy (referring to similar contexts), what he calls generalizing through analogy. This applies to external generalizability to other groups and companies (cf. [45]). Internal generalizability (generalizable within a group, see [45]) has been supported by the use of source triangulation and using two similar teams identifying improvements independently.

In order to allow others to judge the degree of usability and usefulness, researchers have taken care in describing as many contextual elements as possible using the checklist provided in [32] as a guide. In addition, the VSM framework is designed providing alternatives of how to implement different activities. This makes the value stream map flexible.

Reliability – interpretation of the data is influenced by the background of the researcher/repetition of the study: Whenever conducting studies with a vast amount of qualitative information, there is a risk that the interpretation of the data is influenced by the background of the researcher. The risk cannot be mitigated, but only reduced. In order to reduce this threat, the following actions have been taken: (1) Two researchers were involved in the interpretation of the notes and documentation; (2) the documented result was reviewed by a colleague at the company; and (3) the colleague at the company was also taking notes during the workshop allowing for a comparison of notes. In addition, member checking with the whole VSM team has been conducted during the retrospective activity in the value stream process.

Internal validity was not considered a threat in this study as the researchers do not aim at forming a statistical causal relationship between variables (this threat is mainly considered in experimental/sampling studies [46]).

7. CASE STUDY RESULTS

The case study results are split into two main parts. The first part illustrates how the value stream framework is instantiated in the specific case and provides arguments of why in the given context certain decisions for instantiations were made. In the second part, the actual outcome in the form of current state map, future state map and retrospective are presented.

7.1. *Instantiation of the framework*

The management of the case study company agreed to commit 12 persons to the value stream map activity for approximately 10h (including the steps mapping the current value stream map, identifying waste, improving the process and performing retrospective analysis), which adds up to 120 person hours. The selection of strategies and the distribution of time between steps were performed under these constraints.

7.1.1. S1: Initiation. As outlined in the framework, the steps for getting started are the selection of key stakeholders, determining the purpose, defining the team, training the team, scoping the problem, defining the value and understanding value creation.

S1.A1 Identifying key stakeholders: Three alternative options are suggested, namely direct internal customers, indirect internal customers and external customers (see S1.A1 in Appendix A). As the company is operating in a market-driven context where a product is developed for a mass market

with many potential customers, no external customer was invited to participate. If the company would have been in a bespoke and contract-based situation, the external customer would be more easily accessible. Hence, only direct internal customers and indirect internal customers are included. In the case company, direct internal customers are people working with the release and packaging of the software product. Indirect internal customers are participants of the process, such as people responsible for design and testing.

S1.A2 Determining the purpose: During the planning phase, the purpose was to evaluate a concrete improvement. The reason for evaluating a concrete improvement and not focusing on the overall process was to not flood the organization with too many new improvement initiatives, which would be a risk for the ongoing improvements. The concrete improvement to be evaluated was focused on restructuring the communication between product managers, requirements engineers and development teams in order to facilitate frequent exchange of information and feedback. This was changed after activity S2.A4 as the analysis showed that the improvement evaluated did not target the wastes that received the highest priorities by the team. In consequence, it was decided to change the purpose to optimize end-to-end process toward a specific target with respect to lead time. It was made explicit that the ongoing improvements could be incorporated when improving the current state process, but at the same time new improvements can be proposed.

The workshop participants received the following instructions for the VSM activity:

Your task is to redraw a map that cuts the lead-time down to X days. Here you should implement improvements (e.g. the suggestions related to improved communication between product managers, requirements engineers, and development teams) and determine what changes should be done in order to achieve a shorter lead-time. Keep in mind, however, that later you will evaluate the process based on other attributes (e.g. perceived quality by the customer, cost, etc.), which might be negatively affected. Also keep in mind whether the root-causes for waste are addressed. If you cannot achieve the X days without compromising other important value considerations (see value components), determine what would be a realistic improvement.

In short, this translates into the following goal that should be achieved: ‘Reduce the lead-time by X% while developing usable and useful features for the customers, without defects, as efficiently as possible, and at the right time.’

S1.A3 Defining the team: Whom to include in the team was very much restricted by the resources committed to the value stream map. The company decided to focus on two systems (systems A and B) that are part of a larger system (see description of S1.A3). With the focus on the two systems, each system should have at least one representative in each process area (product portfolio, system responsibility, and system integration and release). Only chefs were included in the first value stream map initiative. The chefs were selected based on their experience in the process, that is, they all had very good knowledge of the processes currently practiced. Also process and product owners should be present within each step. A process owner is responsible for driving process improvements. A product owner is a main stakeholder having requirements on and an interest in the product. The set-up of the team is summarized in Table IV.

Table IV. Value stream mapping team.

Process areas	Roles	Participants
Product portfolio and requirements	Product and system management, program manager	3 product owners, 1 process owner
System design, function testing and system level test	Architects and designers, coders, program manager	2 product owners, 2 process owners
System of systems integration (compound system test) and release	System level test testers, program manager, release and product packaging	2 product owners, 2 process owners

S1.A4 Training the team: The teams were trained through a Powerpoint presentation containing the following information: (1) The goals of VSM and the potential benefits illustrated through the case study presented in the MIT VSM manual; (2) an overview of the VSM process; (3) To-Dos related to the current state map; (4) notations of VSM; (5) To-Dos related to the future state map; and (6) presentation of the VCs with examples. After the presentation, there was a question and answer session where the content of the presentation was clarified.

S1.A5 Scoping the problem: Solely focusing on documentation analysis was not considered as opinions (through discussions) are needed to understand why specific problem bounds should be defined. Moreover, it is very important to involve key persons in the discussion who can champion the VSM in order to gather the required competence needed for the initiative. Collecting as much information about the existing process before going into an interview is recommended, because the discussion can be focused on scoping the problem.

Workshops were not chosen as they require too much effort early on in the mapping process. Therefore, documentation analysis and discussion was used as a trade-offs between documentation analyses and interviews. From the documentation analysis, the researchers identified the main process steps. The following decisions were made regarding the problem bounds with respect to the critical elements defined in the framework (see description of S1.5):

End-to-end process: The process starts with the requirements gathering and definition activity, and ends with the system verification activity. The reason for choosing the end-to-end process is the observation that different parts of the process influence each other and hence considering only one activity might lead to suboptimization. For example, a change in the requirements activity could have a significant impact on testing and product packaging activities.

Products: The product that was chosen is an SoS. As described in the research method (Section 7.1), the company is employing an SoS development approach. In total, nine systems are parts of the overall system. The most critical systems were selected for the mapping activity, that is, a delay in the development of systems A and B has the most significant effect on the development of the overall system. This is due to the size of the systems and the number of dependencies to the other systems.

Product owner: For each of the process areas, a process and product owner have been identified (Table IV).

Customer of output: The internal customer is release and product packaging responsible for delivering the verified system and making sure it is installed and configured according to the customers' needs. The external customers are telecommunication operators on the market.

Delivery of input: The high level needs are identified by market units (e.g., through market surveys and customer visits). The market units then communicate with the development units regarding the requirements.

S1.6 Defining the value and understanding value creation: The selection of VCs is driven by the goal question value approach. The goal can be subdivided into several subgoals:

develop useful and usable features;
without defects;
as efficient as possible; and
at the right time.

From these corresponding questions and related VCs were derived using the software value map, shown in Table V. The impact of proposed improvements will be evaluated with respect to the identified VCs.

An overview of the instantiation of the phase 'getting started' can be found in Figure 5.

7.1.2. S2: Mapping the current value stream. As outlined in the framework, the steps for creating the current state value stream map are the zeroth map, identification of tasks and flows, data collection, evaluation of value and understanding of iteration.

S2.A1: Drawing the zeroth map: The principle tasks were retrieved from documentation and measurements and were already well-known to the researchers and the organization. Hence, this step required little investigative effort. The main steps of the zeroth map were defined on the basis of the

Table V. Goals, questions and values.

Subgoals	Questions	Value components
Useful and usable features	Which functions are really needed, and how easy are they to use?	Functionality, quality of delivered product
Without defects	What is the level of quality with respect to reliability, portability and maintainability?	Quality of delivered product
As efficient as possible	Are we developing the product efficiently, and do we have the knowledge and motivation needed to do so?	Internal efficiency, human capital value (please note that these value components are beyond customer perspective)
At the right time	Are we able to deliver and install the system efficiently?	Delivery process value

states that requirements can enter (e.g., being in specification, being in design and being in test) and so forth. An overview of the states can be found in [47].

S2.A2: Identifying tasks and flow: The high level activities of the zeroth map were not further refined for the VSM activity. The motivation was that (1) objective measurements were available for the steps; and (2) in discussion with the practitioners, the researchers came to the agreement that the analysis of the high level steps will reveal the wastes and improvement potentials with respect to value. Furthermore, the map could be further refined when needed (e.g., if the researchers feel that important wastes could not be discovered because of a too high abstraction level). Given the good understanding that the researchers already had of the main flows and tasks, documentation analysis and discussion was sufficient. In other cases where the processes are not documented and measured, more time intensive methods might be required to obtain a more complete picture (such as workshops and/or in-depth interviews, see Appendix B).

S2.A3: Collecting data: Often data in software engineering show a high degree of variance, be it with regard to productivity [48, 49] or with regard to lead times [50]. Hence, few critical requirements have to be chosen that would lead to general improvements in the development process and that can be used to retrieve measurements and set a baseline for the VSM activity. In this case, the requirements chosen should (1) affect multiple systems in the SoS structure as single system requirements are less problematic to develop; (2) the requirements should be of medium size according to the interval set by the company. Three requirements fulfilling these criteria were selected and based on these requirements an initial draft of the value stream map was prepared including actual measurements. The requirements were selected based on documentation/measurements and discussion with the practitioners at the company. After having an initial map with tasks, flow and measurement data, a workshop was conducted to finalize the current value stream map.

S2.A4 is described in S3 as these activities were combined.

S2.A5: Understanding iteration: An approach for understanding iteration was not implemented in this value stream map because the zeroth map was used as an input for the activity, given that measurements were available on that abstraction level. However, this does not mean that challenges related to iteration will not be discovered. In fact, several root causes related to iterative behavior in the current value stream map were identified (see Section 8.2 – loopbacks, test cycles and rework iterations/steps).

An overview of the instantiation of the phase ‘mapping the current value stream’ can be found in Figure 6.

7.1.3. S3: Waste identification. This step was combined with S2.A4 (evaluation of value). The reason being that all activities contribute some value (either external or internal). For example, everyone would argue that testing is valuable. Hence, in this case, the researchers found it is more suitable to start focusing on improvement potential based on long waiting/lead times. The workshop as an approach was selected as people in different roles should interact with each other and provide immediate feedback. This could not be achieved with any other alternative provided (e.g. individual interviews and study of documentation). Furthermore, agreements and disagreements are revealed immediately and can be resolved. In the end of the current value stream map activity, it is important that everyone in the team has the same understanding of the current state map and agrees on the

Step 2: Mapping the Current Value Stream Map	
S2.1 The Zeroth Map	The Zeroth map was created based on the high-level process the requirements flow through, identified in the documentation. The process was already well understood by the persons organizing the value stream map.
S2.2 Tasks and flow	☐ Documentation analysis ☐ Interviews ☒ Documentation analysis and discussion ☐ Workshop with team members ☐ Follow the work
S2.3 Data collection	☒ Documentation analysis ☐ Interviews ☐ Documentation analysis and discussion ☒ Workshop with team members ☐ Follow the work
S2.4 Evaluation of value	The evaluation of value was done with S2.3
S2.5 Understanding the iteration	N/A
Step 3: Identifying Waste	
S3: Waste identification	☐ Documentation analysis ☐ Interviews ☐ Documentation analysis and discussion ☒ Workshop with team members ☐ Follow the work Note: S3 was combined with S2.3 in a single workshop.

Figure 6. Instantiation of the phases ‘mapping the current value stream’ and ‘identifying the waste’.

outcome. This is important as the current state map serves as the reference for creating the future state map. In the workshop, the team was asked to perform the following activities:

Review data collection: Review the prepared times (lead times, WTs and PTs) and change them according to their experience. (30 min)

Identify wastes: Identify parts in the process with the highest improvement potential; For example, in which phase do we see the majority of waiting, where are the activities that take very long time? This was performed by putting burst signals within the value stream map containing the updated and agreed upon lead and WTs. The burst signals were then prioritized by the team. For prioritization, cumulative voting was used as this is known to be a fast and accurate method to prioritize items [51]. (30 min)

Identify reasons for wastes: Identify the reasons/root causes for long WTs/PTs and prioritize the root causes. This was achieved by having each participant write the main root cause for each of the burst signals on a yellow sticker. The moderator (second author of the paper) collected the yellow stickers and clustered them with the help of the workshop participants. (45 min)

Information about the next steps. (5 min)

The practitioners received a presentation of the value dimensions to make sure that when they decide on what is waste, they have a clear picture of the dimensions. What might be waste from one dimension (example: ‘waiting to allow for a second thought as desired outcome’ in Table VII might be wasteful from the internal efficiency – time perspective in Appendix D B.3) may not be from another (e.g., the external customer perspective in the example as practitioners need time to think, reflect and clarify to really deliver the functions the customer needs). Later, the dimensions were used for a rating of the effect of the improvement implementation on the values, combined with the practitioners providing rationales. This also relates strongly to defining waste beyond the customer perspective, reflected in the title of this paper, as waste as well as improvements are looked at from different perspectives, hence extending the view of what is used in previous value stream analyses (based on the well-known seven wastes presented in, for example, [5]).

7.1.4. S4: Process improvement. This step contains of three activities: (S4.A1) identify strategies to eliminate the waste, (S4.A2) draw the future state map, and (S4.A3) to reevaluate value. S4.A1 and S4.A2 were combined in a single workshop. Other alternatives would have been to conduct document analysis, interviews, or a combination of document analysis and discussions. However, for the same reason presented in S3 (Section 8.1.3), workshops were conducted, that is, to allow for interaction and immediate feedback between team members. The workshop was structured as follows:

Introduction: Presentation of the results from the first workshop (i.e., current state map and root causes) was given to the participants to refresh their memory of the results. Furthermore, the VCs were presented again, as this need to be considered when proposing the improvements. (30 min)

Identify improvements and redraw map: The participants were to identify improvements that can tackle the root causes and to redraw the process based on the improvements identified. Ongoing

Table VII. Root causes versus activities.

General cause	High-level requirement (HL-Req)	Commit for development	System level test/compound system test	Write RS
Many parallel items	Many HL-req. in development in parallel; too high ambition in the beginning, ending up with less in the end	X	Too many deliveries shortly before release intent and late system integration	X
Level of detail	High quality early for good estimates, too high ambition (document up to 80 pages)	Discuss too much detail delaying decisions	X	Specify solutions, not requirements; different expectations on level of detail
Involve too many/wrong parties / communication	Estimations/solutions require many people; document/feedback-oriented negotiation	Waiting for clarifications	X	Lack of communication /cooperation between requirements, implementation, test; long time for handovers
Change (ability to change, request of change)	New input from customer	Waiting to allow for second thoughts as desired outcome not clear enough	X	Come up with new ideas/solutions due to late involvement of design
Size	X	Too large requirements to fit into release scope (cannot approve parts of requirements)	X	X
Release focus	X	Try to fit requirements into certain release instead of starting to work early; cannot commit resources on parts of requirements	X	X
Loopbacks	X	X	Repeated fix/correct cycle for internal faults	X
Length of test cycles	X	X	Waiting for final delivery before starting of test; too many test cycles	X
Many rework steps	X	X	Long time due to many activities – understanding problem, generating new package, reverifying	X
Quality of tests/test efficiency	X	X	Too big fault-slip from systems to compound system integration	X
System focus	X	X	System focus performs well, challenges in integration	X
Key performance indicator influenced process	X	Key performance indicators drive wrong behavior (wait too long with starting in development)	performance at compound level X	X

improvements (if there are any) should be taken into consideration. As the number of workshop participants was 12, there was a risk in having too many people in one workshop, which could lead to the likely outcome that the results could not be obtained in time. Therefore, the team was split into two subteams, making sure that each role is represented in each subteam. Furthermore, before the teams were split, it was stressed that the teams should consider medium sized requirements under realistic assumptions, that is, that there is much ongoing work in parallel and that there is high market pressure with respect to lead time and quality. That is, the new scenario should not be over simplified or based on 'best case' assumptions. For redrawing the map, the teams were given 90 min. When the time was up, each subteam presented their map, and a comparison /consolidation of the maps was performed by agreeing on the improvements identified.

Reevaluate value: For the reevaluation of the value, each individual filled in a form answering the degree of value gain/loss based on the improvements. The scales in which value gain/loss was evaluated are presented in Appendix B. Furthermore, each team member was asked to write down why they thought the identified improvements impact value the way they answered.

An overview of the instantiation of the phase 'improve the process' can be found in Figure 7.

7.1.5. S5: Retrospective analysis. The retrospective was conducted in two steps. In the first step, a walk-through of the results was performed. In order to conduct the walk-through, the moderators prepared a written report describing the results obtained in the previous steps. Walk-throughs are particularly well-suited to inform a group of people about the result and to collect feedback. During the walk-through, every participant received a printed version of the report, and the moderator went through the report page by page explaining the contents of each page. The participants were allowed to interrupt and ask questions at any time of the walk-through.

In the second step, the researchers collected feedback on the VSM process asking three questions as follows:

What was good about the process?

What was not so good and how can it be improved?

Would you like to use VSM as a tool for improvement in the future? Rate on a scale from 1 (no, never again) to 10 (should be a continuously used method for improvement). A value of 5 means you are indifferent.

An overview of the instantiation of the phase 'perform retrospective' can be found in Figure 8.

7.2. Outcome of framework application

Four main work products were produced during the value stream activity as follows:

Current state map including zeroth map, burst signals indicating improvement potential due to undesired behavior.

Wastes and root causes for the burst signals (Table VI and Table VII).

Step 4: Improve the Process	
S4.1 Eliminate waste	☐ Documentation analysis ☐ Interviews ☐ Documentation analysis and discussion ☒ Workshop with team members
S4.2 Draw future state map	☐ Documentation analysis ☐ Interviews ☐ Documentation analysis and discussion ☒ Workshop with team members Note: S4.1 was combined with S4.2 in a single workshop
S4.3 Re-evaluation of value	Survey was handed out at the end of the workshop focusing on steps S4.1 and S4.2.

Figure 7. Instantiation of the phase 'improve the process'.

Step 5: Perform Retrospective Analysis	
S5 Retrospective	☐ Informal review ☒ Walkthrough ☐ Technical review ☐ Inspection

Figure 8. Instantiation of the phase 'perform retrospective'.

Table VI. Prioritization of bursts.

Alarm	Description	Rank (votes)
High-level requirement	Overall long time with 50% waiting	1 (41 votes)
Commit for development	Mainly waiting	2 (30 votes)
System level test/compound system test	Mainly waiting	3 (22 votes)
Write RS	Overall long time with 50% waiting	4 (16 votes)

Future state map including improvement proposals, a future state map indicating changes based on improvement proposals, and estimates for new lead and WTs. Furthermore, the reevaluation of value was conducted.

Retrospective containing reflections on the value stream process and the delivered result. In the following, a narrative of the work products and the reflections provided by the workshop participants are given.

7.2.1. Current state map.

Review data collection/zeroth map: The workshop started with a presentation of the zeroth map that was drafted by the researchers prior to the workshop. The map is shown in Figure 9. It contains the following activities:

HL-Req: The requirement is specified such that the development organization has an idea of the customer's needs, and estimations for time can be made. This activity maps to high level specification (prio) in Figure 4.

Commit for dev: At this point, a decision is made whether the development organization should commit development resources in order to implement the requirement. The answer has to be 'yes' in order for the requirement to enter the 'compound system development' box (in Figure 4).

Write requirements specification (RS): The detailed requirement is specified, so that it is precise enough to be developed in system development, this activity maps to the detailed requirements specification (DRS) activity in Figure 4.

Design and function testing (FT): The requirement is implemented in system development and tested within the teams (Agile Team Sprint in Figure 4).

LSV/C-LSV: The integrated result of the system development is tested at the LSV. The compound system is tested in the C-LSV.

During the workshop, the team filled in the lead times and WTs within the activity in percentages, as these were not available as data prior to the workshop. Furthermore, some of the WTs in-between activities were moved into the activities, as the participants argued that there is some overlap between the phases, and that items generally do not get stuck in between the activities. Given the description of the activities after 37% of the lead time, development resources are committed to development. After 82% of the lead time, the first verifiable version of the requirement is completed, and after 100% of the lead time, a verified and usable feature is ready to be made available to the customer.

Identify wastes: Based on the lead time/WT distribution, the burst signals were placed, indicating improvement potential/need for improvement in terms of overall process lead time or WT. The

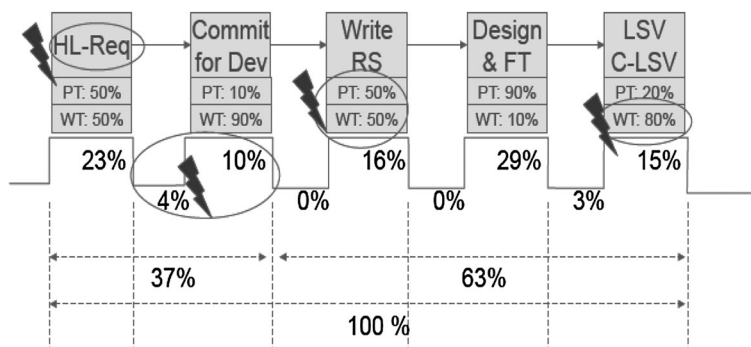


Figure 9. Zeroth map current state.

activity of drawing the zeroth map was finalized after everyone agreed to continue with the values on the whiteboard. In order to focus the improvement effort, the bursts were prioritized using cumulative voting. The overall ranking was obtained by adding up points of individual participants for each phase. The ranking is shown in Table VI.

Identify reasons for waste: Based on the clustering of the issues in the workshop, a matrix was obtained showing general issues and their mapping to the different process activities. The root causes helped the participants to target the right issues when searching for improvements. As can be seen from Figure 5, the 'HL-Req' specification has the highest improvement potential, based on the opinions of team members, followed by 'commit for development', 'LSV/C-LSV' and 'write RS'. Figure 6 illustrates the result obtained from the workshop. What should be observed here is that a number of general causes have been identified (such as many parallel items). Some of the general causes apply to several phases, whereas others are unique to a specific phase. The causes are not elaborated in further detail; the purpose of the table is to provide a general overview of what type of causes have been identified to demonstrate the capability of the value stream approach to lead to a good coverage of causes across the life cycle.

7.2.2. Future state map. To draw future state map, the team was again split into two groups. Each group returned with very similar improvement ideas and easily agreed on a combined list of improvements. The groups only differed with respect to visualization of the results. The improvement ideas are listed in the following text.

The level of detail in which the HL-Req is presented should be drastically reduced. The HL-Req should be one pager, which should be used as an input to define user stories (i.e., an HL-Req is sliced into several high-level user stories – from here on called epic user stories). The improvement addresses the cause of 'level of detail' (Figure 6).

Integrate value (epic user stories) early: One of the challenges identified was that parts of a HL-Req could not be integrated and verified early on. This was partially due to the way requirements were committed to the development organization (i.e., not parts of a HL-Req. could be committed) and the release focus (Figure 16). As a solution, an HL-Req should be split into epic user stories that are (1) verifiable and (2) deliverable. That is, when defining the requirements, it is important to find a good modularization in terms of features for each HL-Req. An epic user story then can be verified more independently and be useful to the customer, even when some epic user stories related to the requirement are not yet implemented.

Communication and interaction: People from different roles in the organization should be put together to allow for better communication and understanding. For example, persons responsible for system management and development teams should interact, and there should be frequent interactions so that system managers are aware of what is happening in the team developing their requirements. Also, it is important to assure that people across the overall development life cycle interact, which means involving release and product packaging teams earlier and merging system managers, program managers and development teams to communicate frequently (e.g., by co-locating and assure frequent possibilities for interactions).

Handover and documentation: The amount of handovers between system management and development in parallel should be reduced to increase direct communication and interaction (face-to-face interaction and communication replaces/reduces/simplifies documentation).

Increase parallelization: Parallelization between systems and within a system should be increased (e.g., in order to avoid systems waiting for each other). However, it is not clear how to achieve this. One of the challenges is to allow for backward comparability to always be able to integrate with another system in the SoS structure. For example, if system A is updated to a latest version, and system B is not updated yet, a system integration cannot be run because of lack of compatibility (as system A is not backward compatible).

Inspection/Review: Inspections and reviews of each other's documentation should be reduced and direct communication across roles should be used instead (related to items 2 and 3).

Welcome change of plans: Resources for epic user stories should be reserved in plans, and the progress should be continuously reflected upon. Based on the reflection, flexibility is needed when it comes to re-planning to have a higher return on investment (re-planning for value).

Retrospective: Continuous reflection on work processes and plans should be carried out to learn and improve the ways of working.

Team interaction: Team members of different systems should be combined to achieve a system/solution perspective of the overall SoS instead of just having a single system perspective.

Prototyping: When useful, prototyping should be carried out to achieve a better understanding of what is needed by the customers.

What set the teams apart was that one team introduced prototyping as a solution. Otherwise, both teams were in-line with their improvement proposals. However, when it comes to visualizing the solutions, both teams have drawn different pictures. The pictures have some notable differences, but at the same time share the ideas of the improvements listed before.

Result of Group 1: The new process map by Group 1 is shown in Figure 10. The left upper box represents the new HL-Reqs document (QSM 1 pager) that is reduced in complexity. There is an estimated WT in between the completion of the QSM 1 pager and the start of development. The prerequisite for starting is to take a so-called 'Go' decision on an epic user story, indicating that resources of the development organization should be committed to the implementation of the epic user story. When this is done, the epic user story is specified and broken down in development user stories. These are then developed and function tested, and thereafter system tested and compound system tested. According to the team system testing and compound system testing can be parallelized to a high degree. What is also visible is that activities are ongoing in parallel. For example, whereas the third epic user story is specified, the user story from the second epic is specified, and the user story of the first epic is implemented and function tested. Looking at the time lines, the following improvements are estimated, as shown in Table VIII. The table shows the percent of lead time needed to complete certain goals. The first improvement is related to when the organization is able to commit the resources for an HL-Req of medium size. The next row shows when the first verifiable unit to test is delivered. In the current state map, this was at 82% of the overall lead-time, whereas in the new process, it is only 22.50% of the overall lead-time needed in the current state map. The reason is that the HL-Req is now split into verifiable subunits (called epic US slice). Hence, the first verified and deliverable unit is ready after 38.10% of the old lead-time, whereas the first verifiable and delivered unit in the old process could only be ready at 100.00% of the lead time. To complete the same amount of functionality (the estimate of the team was that a medium-sized HL-Req could be broken down to approximately five epic user stories) 50% of the old lead-time

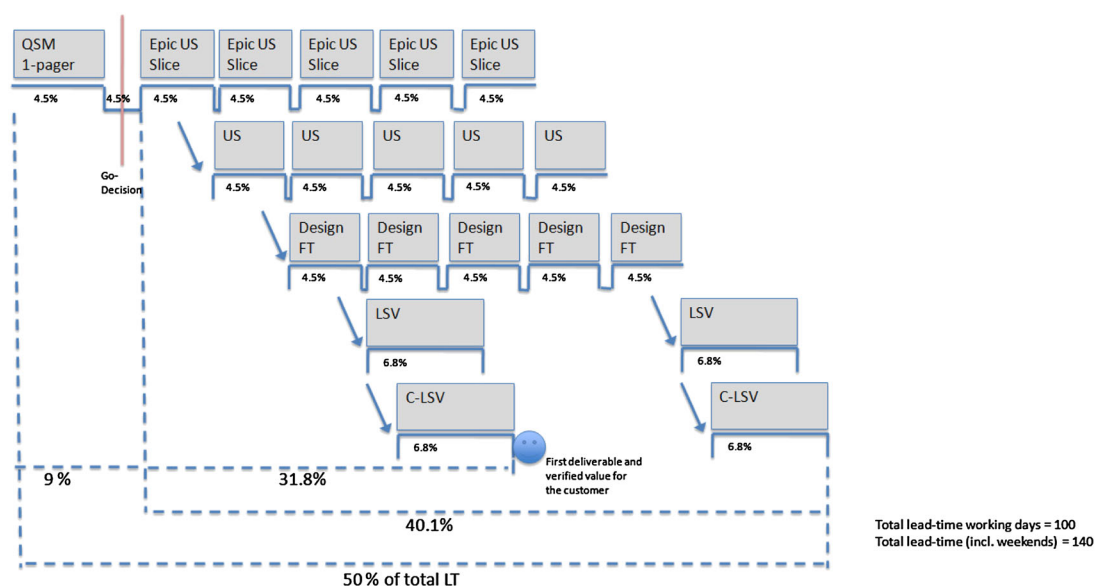


Figure 10. Future state map Group 1.

Table VIII. Improvement in percent.

Lead time	Current	Future	Improvement
Time to commit resources	23.00	9.00	14.00
Time to implement first verifiable unit for test	82.00	22.50	59.50
Time to complete test of first verifiable/deliverable unit	100.00	38.10	61.90
Time to complete the same amount of functionality	100.00	50.00	50.00
Ratio waiting time/productive time	47.40	2.25	45.15

would be needed, given the high degree of parallelism. With regard to WT, the estimations were extremely optimistic, going down to 2.25% from 47.40%.

Result of Group 2: The results of Group 2 look very similar to the results of Group 1; they are just expressed in a different way (see Figure 11). Both maps show parallelism with regard to requirements work, implementation and testing. The difference is that the Group 2 has expressed the overlap in roles in the visualization. Furthermore, prototyping as a concept has been stressed. This group also stressed that value should be created continuously. Thereby, they separated value in terms of function and quality, according to the definition of VCs in Appendix B. The value growth of the current in comparison with the future state map is illustrated in Figure 12. The figure shows that currently, features are integrated quite late in the development process, and quality assurance has no other option than following function integration in time. This is very much in line with the observation in the current state map, where the first verifiable unit is delivered to test at 82.00% of the lead-time. In the new state with the improved process, features are integrated continuously, which also allow for continuous quality assurance.

Furthermore, in the new situation, the curve is not perfectly linear, as illustrated in Figure 12. So far, the reevaluation has mainly been focused on time, but the participants also evaluated the impact of the set of improvements on the VCs.

The evaluation, by the participants, of the proposed improvements with regard to the VC ‘external customer value’ is shown in Figure 13. The figure shows a high agreement that functionality and quality are positively influenced; however, there is some doubt with regard to the effect on external quality.

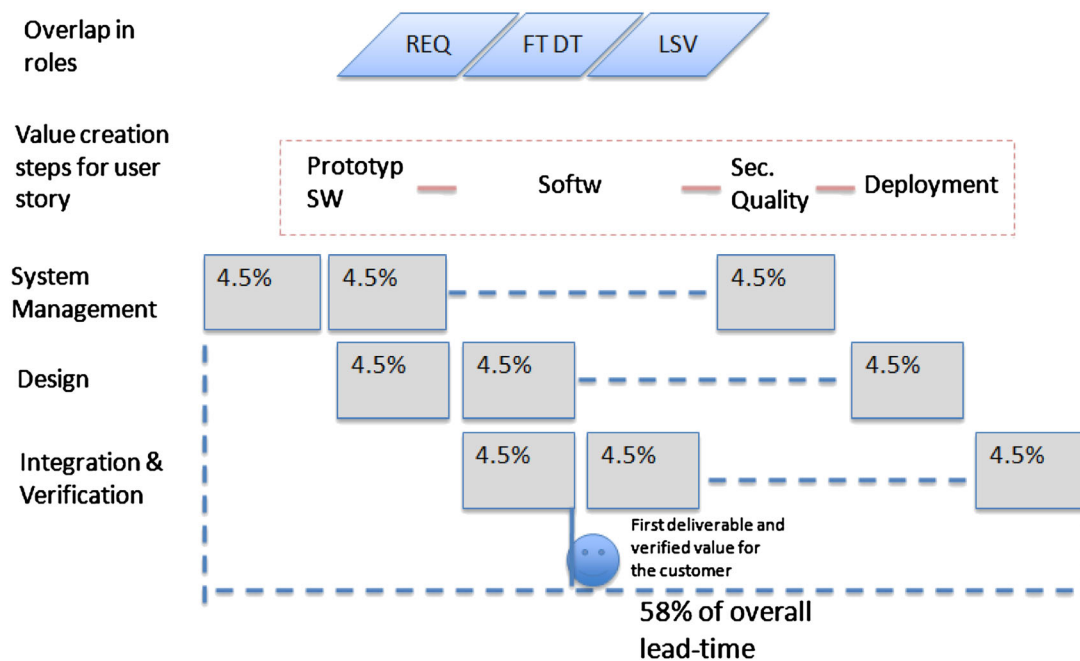


Figure 11. Future state map Group 2.

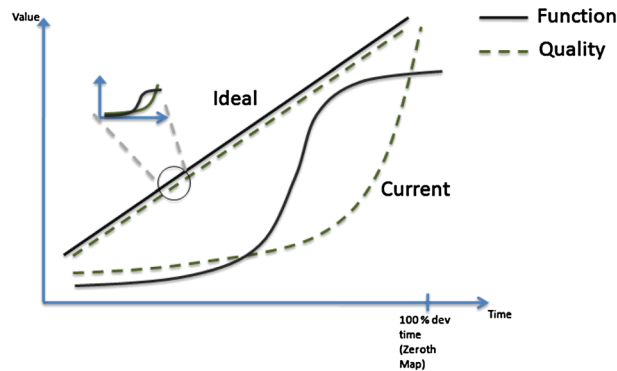


Figure 12. Growth of value over time (current vs. future).

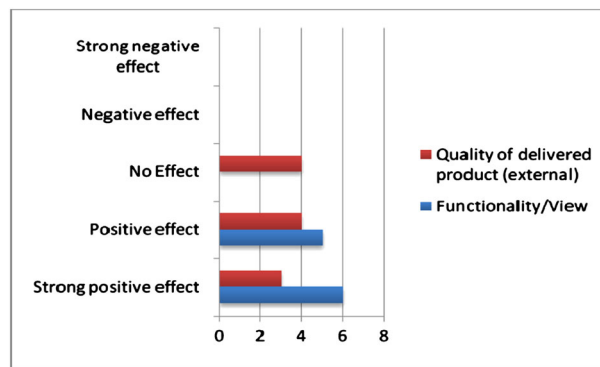


Figure 13. Value component: external customer value.

Positive effect: For the positive effect on functionality, the participants provided the following reasons. First, the new approach increases flexibility by splitting up requirements into epic user stories that are deliverable and verifiable (i.e., requirements are split based on value considerations). The epic user stories also allow for earlier feedback of whether the right scope is delivered. Functionality is delivered more accurately as the company is not implementing what has been decided a long time ago, as (given the dynamic market) this would not be what the customers want today. Also, the issue of parallel items is addressed as not everything is performed at once, and only parts of the functionality can be delivered. No negative effects have been identified for functionality. With respect to quality positive effects have been identified. Due to more frequent deliveries, quality is assured more often. Furthermore, the reliability is said to be improved due to shorter feedback cycles, and hence smaller changes have to be delivered to test. Fewer defects being reported will also increase maintainability. As can be seen in Figure 13, four persons think that there is no positive effect on quality with respect to usability. Their reasoning was that usability is not affected as improving usability requires a higher involvement of the real customer (e.g., through participatory design and creating mockups with the customer). Furthermore, the requirements elicitation should move more from a function centric to a quality centric dialog with the customers.

The evaluation, by the participants, of the proposed improvements with regard to the VC 'internal efficiency' is shown in Figure 14. Disagreement is visible regarding the impact on cost.

Positive: Most people agree on positive impact on time and quality; however, there is some doubt that there is an effect on time, cost and quality. Furthermore, three persons saw a negative effect on cost. The positive effect on time was explained by having higher parallelization right from the start, meaning not so much time spent on guessing but rather doing and delivering, and that the focus on demos and working software will provide results earlier. With respect to cost, few persons suspected positive effects, but did not provide an explanation. The positive effect on quality of work products follows the same argumentation as for external quality.

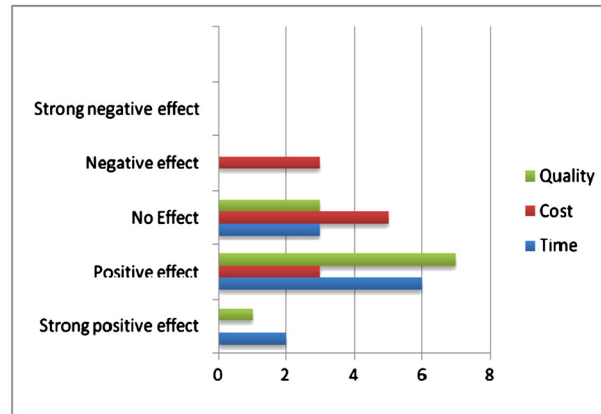


Figure 14. Value component: internal efficiency (beyond customer perspective).

No effect: The figure shows that several participants said that there is no effect on time/cost/quality. For time, they suspected that it will probably be the same, the only difference being the content is delivered much more accurately (i.e., they felt that the estimations were overoptimistic). They also had the feeling that there will be changes in several areas, but overall the time might be the same. For cost, some suspected that the cost is just moved from one point to another, which some also believed is true for software quality.

Negative: People suspecting a negative effect on cost mainly explained this with the increased cost of cross-functionality, as more resources have to be involved at the same time, such as system management has to support people doing implementation, which would require more people filling in different roles.

The evaluation, by the participants, of the proposed improvements with regard to the VC 'delivery process efficiency' is shown in Figure 15. There is disagreement with regard to the effect on the delivery process; overall the majority thinks that there is no effect.

Positive Effect: A positive effect in quality can be seen since by frequent installations and rollouts; the company can learn and become better in delivering upgrades. Also, the quality in the delivery process is improved with product quality. A positive effect on time was argued by saying that the releases would be delivered faster if product packaging and release are involved earlier, as then they can prepare in parallel to development. Furthermore, partial deliveries will be possible with the new process. With respect to cost, a positive effect is seen because better quality of the system will ease rollout and installation during deliveries.

No effect: On the other hand, people saying there is no effect on time argue that it still takes time to have the large systems running at the customer site given that each customer has to retest in their own environments. Furthermore, when not going away from a focus of few major release deadlines

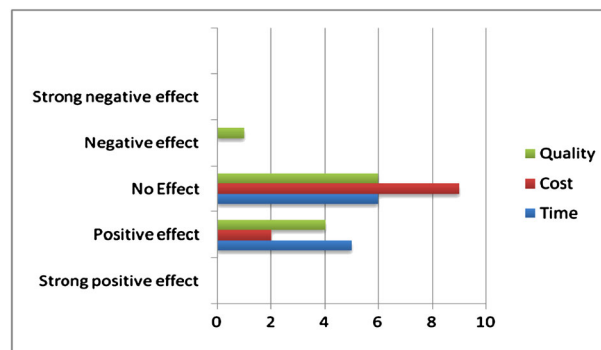


Figure 15. Value component: delivery process efficiency.

per year, there will be no change as homework behavior can be observed, that is, work is performed more intensively closer to deadlines. In other words, if we ought to deliver continuously work will be performed more continuously over time. The reason of few major releases was also raised with respect to no change in cost as the size and complexity of the product delivered to product packaging would still be the same. No effect on quality was seen due to that there is still a focus on major releases, which is not seen as a good driver for frequent verification/delivery of intermediate value. *Negative effect:* The person stating that the changes have a negative effect was saying that it is a gut feeling, not having a good explanation for the reasons.

The evaluation, by the participants, of the proposed improvements with regard to the VC 'human capital value' is shown in Figure 16. The figure shows that there is a high level of agreement on strong positive effects on competence, learning and intellectual agility.

Positive effect: The reason reported for increased competence and intellectual agility was that cross-functional work teams will contribute positively to competence development. However, the participants suspected that the competence would change to having a broader knowledge, but not necessarily deeper knowledge in a specific area. For increase in motivation, many different reasons have been reported. For example, people will be more motivated by knowing the reason of why they are doing a specific task, the reasons being known due to interactions between different roles. That is, a designer and a tester will see a market value of what he/she is producing. The motivation also increases when everyone can see the whole chain/value flow. Early feedback and understanding of features and functionality will also be a motivator. Finally, the new suggested way of working will challenge employees in a new way (e.g., with respect to formulating HL-Reqs in one page and then communicate instead) and hence will make working more fun. The summary of the results of the value reevaluation shows that the survey allowed having a balanced view of the impact of the improvements on value.

7.2.3. Retrospective. During the retrospective the moderator walked the participants through the report. Afterwards, an open discussion followed. The discussion was based on the way forward. That is, what obstacles and open questions are there today in order to achieve the new process, and what can be performed to remove the obstacles and answer the questions. Based on the discussion, four key questions have to be answered:

How do we change our ways of estimating and planning?

Which variable level of detail should we aim for?

How do we synchronize in a good way, and what is a reasonable level of synchronization/unsynchronization?

How do we align our efforts with development taking place in India?

For the questions, the following answers have been provided:

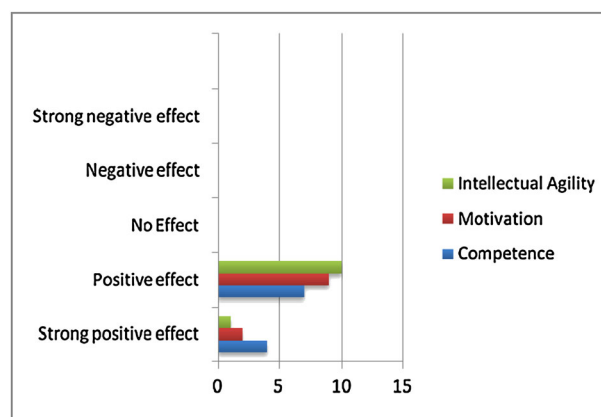


Figure 16. Value component: human capital value (beyond customer perspective).

Planning was performed on the basis of major release intent. However, in the new situation where more flexibility is possible and not so much details are available, planning has to change as well (e.g., how to handle estimations). This also raises the question as how to handle release planning and road mapping of features. A solution for this question has not been provided at this stage.

This question is related on how to obtain development started within 9% of the lead time being spent on HL-Reqs. There might be different needs with respect to detail (some might need more detail, whereas others need less). One possible solution was provided by looking at company internal success stories. Experiences from one product showed good results with prototyping, which means that full picture is not needed before starting. Moreover, requirements stated by product managers in an understandable and precise manner also help to start working. Currently, requirements are often one liners, but there are good examples from other systems recently as how to write HL-Reqs in an understandable way. A way forward might be to find these 'model' requirements as a guide for improvements (e.g., good use cases). Complementary, workshops with software product managers and developers early on can also be useful.

In order to achieve shorter lead times in the early phase (before the decision of committing development resources), systems need to be able to work more out of sync. However, too much out of sync causes a risk when requirements change, and the systems do not work as a whole after integration. A possible solution is that one system takes lead in developing the core, and other systems catch up building on the core. Furthermore, the degree of dependencies between nodes highly depends on the handling of interfaces and network protocols in order to allow for backward compatibility.

Improvement efforts should be aligned with improvement efforts carried out in India. A possible solution proposed was to conduct a similar value stream activity with development sites in India to identify their idea of process improvements focusing on value and waste. The proposition/gut feeling is that when drawing the future state map India would like to have the technical details in place before starting.

After having discussed the way forward, the participants assessed the overall value stream activity saying what went well, what can be improved, and whether they would like to have VSM implemented as a process improvement method. The following points have been raised as being positive throughout the value stream activity:

Very good discussions in the workshop.

Every part of the organization was involved to exchange views and learned from each other.

Good reporting/documentation of the value stream activity.

The activity helped everyone to understand the way of working/processes end-to-end.

Pre-meetings and trainings were good to mitigate the risk of not discussing the issues at hand in the workshops.

Everyone participated in the workshops after being invited.

The view and consideration of different values.

The end-to-end focus.

The following points have been raised as being not so good throughout the value stream activity:

Only chefs, not doers in the workshop.

The VSM used in the start was not a good representation (did not show overlaps between activities), the requirements were not a common case, and the most complicated case was selected, but these shortcomings were overcome in the workshops.

The process is perceived as being too academic.

Time intensive to conduct (but, as commented: if actions follow, very much worth it).

The solutions were quite black and white and oversimplified.

Three improvement proposals have been provided:

Shorter time and fewer people versus involve India and also include technical people.

Clearer expectations on the outcome.

Present goals of the workshops more clearly to everyone.

The final step in the VSM activity was to capture the attitude of the team toward the overall activity. The results are shown in Table IX. The table shows that the majority thinks the workshop was very useful. Only a few doubted the usefulness. Observe that not all persons (9 out of 12) provided a vote, as they would make their vote dependent on the support in implementing all the proposals made. In other words, they believe that the improvements identified have a positive effect, but their identification is only worth when they are implemented. A comment from one of the participants was that if the proposed improvements are implemented, the workshop would have a 10, otherwise a 0.

Given the results presented in the case study, the researchers provide a number of recommendations/lessons learned, also referring to the evidence based on which the recommendations are provided.

8. DISCUSSION AND LESSONS LEARNED

In the following, a number of lessons learned from conducting the VSM activity are presented. For each lesson, the researchers refer to the evidence collected in the case study that supports the lesson.

8.1.1. Use workshops for waste identification (S3), waste elimination (S4.1) and drawing the future state map (S4.3). In the retrospective, fruitful discussions in the workshops were appreciated. Direct interaction was possible, in fact people from different parts of the organization (e.g., release and product packaging) hear people talking about preceding process steps that are of interest for their work, but they have not been able to meet the persons before in order to learn about their process and communicate the knowledge. Given the context of the study (SoS) with many teams and activities, it is important that people are given the possibility to meet and discuss process issues who otherwise do not meet in the daily work, also recognizing that they need to communicate in the future. This was also apparent from the root causes being identified, many being related to sharing knowledge in forms of documentation and communication. Also, improvements proposed stressed face-to-face communication across roles and the preference of direct communication over documentation and review.

In smaller organizations where people work in small/few teams other approaches might be preferred (e.g., interviews) given that overall fewer resources are available, and everyone knows the overall process/project structure well. In order to investigate this, further instantiations of the framework are needed.

8.1.2. Time-box activities in workshops when having very limited time and combine value stream steps. As stated in the beginning of Section 8.1.1, very limited time was available for the value stream activity (in total 10 h). Such situations are likely to occur in practice when a method is piloted and not yet established as an accepted practice. With limited time the researchers found that it is essential to time-box activities in the workshops, the time for the time boxes being stated for each activity in the instantiation in Section 8.1. Thereby, it is important to inform the participants about the time boxes and to make sure that they are kept. The time boxes should be planned with buffers (e.g., add 10 min on each time box) to allow for resolving eventual issues that come up during discussions and require more time. Given the restricted time, some steps were also combined in one workshop. However, the researchers believe that it is important to have time between drawing the current state map and the future state map in order to allow for reflection about the workshop related to the current state map.

8.1.3. Stress complexity and realistic context during the whole activity. In the retrospective (Section 8.2.3), an issue was raised that solutions are black and white and oversimplified. The reason is that while

Table IX. Attitude toward value stream mapping activity.

Scale	Number of persons
Workshops very useful (>8 points on scale)	5
Workshops useful (score 5–7)	2
Workshops not so useful (score < 5)	2

creating the new map, an optimal context seemed to be assumed by the participants (i.e., few dependencies to other systems and ability to plan). Hence, it is always important to stress a realistic context. Therefore, the researchers propose to prepare scenarios for the participants that should be evaluated against the new future state map (e.g., what if a project is not completed in a sprint, what if there are many dependencies to another system and that system is delayed, and so forth). These scenarios could be used to update estimation values. The oversimplification issue has been partially addressed in this value stream activity already in the reassessment of value and the retrospective (Section 8.2.3). In the reevaluation of value, the participants provided possible risks why the improvements might not have an effect, and in the retrospective, they discussed the main obstacles/issues that have to be resolved in order to make the improved process work. The researchers believe this provides important insights in order to succeed in implementing a version of the future process that is close to the one proposed in the future state map.

8.1.4. Allow teams to express processes in their own words. During the initiation, there was a question whether to require a specific notation for the new process or to let people express the process in their own words. The researchers decided for the latter alternative as participants indicated it would be easier to express the ideas and improvements in their own ways. However, the only requirement posed was the times (waiting as well as PTs) could be derived from the new process map, allowing for comparability to the current state map. As is indicated in the two maps produced by the groups, a comparison could be made for a HL-Req of similar size and system impact. Hence, there is no foreseeable reason to prescribe a specific notation. In short, it is important what information is captured and not how. In addition, the maps should be complemented by information regarding discussions that the scribe should document (see previous lesson).

8.1.5. Introduce different value components to avoid suboptimization. In lean development 'seeing the whole' is an important principle to avoid suboptimization of specific aspects of the software process [5, 6]. In the case study reported, suboptimization was avoided in two different ways as follows: (1) the goal of the VSM activity was focused on reducing lead time, but it was given as a restriction that other values (derived from the goals) should not be compromised. In case there is a need for compromise, a realistic new goal for lead-time reduction should be set. (2) At the end of the value stream activity the improvements were evaluated with respect to multiple VCs. As has been revealed in the data, not only the positive aspects were discovered. At the same time risks associated to the improvements were identified, such as an increase in cost (see 'Value revaluation' in Section 8.2.3). Hence, multiple value considerations should always be considered when conducting VSM to make sure that no important value is compromised through improvements. In addition, the researchers recommend capturing the rationales for value revaluation.

8.2. Further observations

8.2.1. Assure cross-functionality and end-to-end focus. The end-to-end focus is generally stressed in the lean methodology (cf. [5, 6, 19]). As can be seen in the results, root causes have been identified for every stage of the process, which is often not possible when only looking at one activity. For example, an undesired behavior such as long waiting in a specific phase does not necessarily have the root cause in the same phase. With respect to improvement proposals, the process was redrawn end to end, which also required the input from all parties during the discussion.

8.2.2. Document the process and products produced during the 'value stream mapping' activity carefully. In the retrospective, the participants appreciated the quality of the reporting (Section 8.2.3). It is central to capture and document the activities carefully as a vast amount of information is produced with respect to root causes, improvements, process charts, reflections and so forth. To be able to later on take actions and spread the knowledge of the results in a very large organization, careful documentation and analysis is the key.

8.2.3. Assure handshaking of goals and expected outcomes as early as possible. From experience with previously held workshops in industry, the researchers know when the goals of an activity are not clear, they are discussed during the workshop; consequently, the actual task is not completed. In

order to focus the effort on the actual value stream activity, the goal of the VSM activity was handshaked with the management, and each participant received written documentation of the goals. Given that the activity was successfully completed within the time boxes, there is a strong indication that the initial effort in defining and agreeing on the goals and expected outcomes of the activity contributed to a complete result from the VSM activity.

8.2.4. Involve chefs and doers. As discussed earlier, the number of participants in the workshop needs to be limited in order to be able to finish discussions in a reasonable time frame. In the ideal situation, everyone should be involved. The researchers completely agree with the observations of the participants that it is important to involve the doers as well. However, in this case this was not possible, since time the effort spent on the activity was already perceived as high (see 'Retrospective' in Section 8.2.3). Hence, in the future it is important to conduct subsequent VSM activities with doers, as well as with stakeholders in India.

8.3. Research agenda

Although we have detailed the lessons learned from the case study, further case studies are needed to test the approach in different contexts (e.g., smaller companies), different domains (e.g., safety critical/ embedded systems) or different development methodologies (e.g., purely plan-driven). Furthermore, longitudinal studies are needed that report on using VSM and seeing the effects after years of usage, looking at suggested and implemented improvements and their effects. In order to have a good starting point, this study lays the foundation.

9. CONCLUSION

This paper presents a framework for conducting VSM in software engineering. The framework outlines the process of the VSM activity and provides alternative strategies of how to conduct each step. A case study has been conducted showing an instantiation of the framework. Based on the application, answers to the research questions are posed as follows:

RQ1: What instantiations of the process do practitioners prefer?

The instantiation of the value stream process was performed for each of the steps (S1: instantiation, S2: current state map, S3: waste identification, S4: process improvement and S5: retrospective). The context of the company influenced the choices of instantiation.

The size of the company developing multiple interaction systems required the use of workshops to allow discussion and interaction between groups of stakeholders that are distant from each other in the regular work activities, but are highly dependent on each other's input (e.g., requirements and support); this was particularly essential during S2 to S4. During S3, an interesting finding was that the practitioners perceived the current state map as a too simple representation of the reality, which was found during the retrospective. The notation does not allow illustrating iterations and overlapping activities in a good way. Hence, a possible solution or improvement in future work could be to represent the map in the form of a dynamic model (e.g., using process simulation). During S4, we learned that practitioners could express themselves more freely when exploring improvement ideas when they could use their own notation to describe the processes.

Furthermore, the type of interaction with the market influenced how the map is instantiated. In the value stream activity it was found challenging to involve customers, given that the company works in a market-driven context. That is, a high number of potential customers play a role, which are not accessible yet. In this situation, people representing the product portfolio and requirements perspective were involved. The limited time available also only allowed involving chefs, and not doers. Thus in future activities, it was considered to also involve the doers who conduct the testing and write the code for the product more.

The value dimensions being considered might change. Companies in a different context (e.g., bespoke development or a different domain such as safety critical software systems) might select and emphasize different VCs when compared with this case. In the instantiation emphasis was given on considering five VCs derived on the basis of the goals of the VSM activity. Choosing multiple VCs was motivated by the intention to not suboptimize.

RQ2: Is the initiated process capable of identifying the current state (waste) and desired future state (improvement) accurately?

The outcome of this question highly relies on the expertise of the practitioners participating in the value stream activity. Two measures suggested in literature are member checking and saturation to assess the elicitation of qualitative information. In the case of value stream mapping, the qualitative information constitutes current state map, waste and future state map incorporating improvements. During member checking, the practitioners committed and agreed to the outcome. Very similar improvements and future state maps were identified by two separate groups of practitioners having similar roles in each group, which indicates that the value stream activity is capable of accurately identifying waste and desired state. Furthermore, the practitioners evaluated the improvement actions positively against external customer value, internal efficiency (quality and time) and human capital value (intellectual agility, motivation, and competence). However, some limitations have been mentioned. In particular, one concern was that the improvements and their effect were perceived as a black and white representation that is oversimplified. One potential solution to address this concern is to evaluate alternatives through demonstrating the behavior of the process in a simulation.

In future work, further evaluations and instantiations of the VSM solution presented in this paper should be conducted. An interesting direction for research is to represent the current and future state map dynamically so that practitioners can play through alternative scenarios using simulation, which could potentially address the issue of the perception of having oversimplified solutions and views on the process.

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AUTHORS' BIOGRAPHIES



Dr Mahvish Khurum is the Managing Director at KMH Solution in England. She received her PhD degree in Strategic Software Engineering in 2011 from Blekinge Institute of Technology, Sweden. Dr. Khurum's research area is Strategic Software Engineering, which touches both software engineering and industrial technology and management. She is also working with practical innovation. Dr Khurum bases her research on challenges identified in industry, then develops solutions in collaboration with industry practitioners, and ultimately validates and tests the solutions in a real industrial setting. She has researched in collaboration with IBM, Qtel, and Ericsson. She is also a member of Software Product Management Organization. (<http://ispma.org/personalmembers/mahvish-khurum/>)



Dr Kai Petersen is a senior researcher at Blekinge Institute of Technology (BTH), Sweden. He received his PhD from BTH in 2010. His research focuses on software processes, software metrics, lean and agile software development, quality assurance, and software security in close collaboration with industry partners. Kai has authored over 40 articles in international journals and conferences, and has been the industry chair of REFSQ 2013.



Dr Tony Gorschek is a Professor of Software Engineering at Blekinge Institute of Technology. He has over 10 years industrial experience not only as a Senior executive consultant and engineer but also as chief architect and product manager. Currently, he manages his own consultancy company, works as a Chief Technology Officer, and serves on several boards in companies developing cutting edge technology and products. His research interests include requirements engineering, technology and product management, process assessment and improvement, quality assurance, and practical innovation. Dr Gorschek bases his research on challenges identified in industry, then develops solutions in collaboration with industry practitioners, and ultimately validates and tests the solutions in a real industrial setting. Industrial partners include, but not limited to IBM, Qtel, Ericsson, Daimler AG, and Volvo Cars. For more information/ publications, contact <http://www.gorschek.com>.

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